

SINKGRAB · REQUIREMENTS

SinkGrab requirements

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DOCUMENT CONTROL

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Status	Draft	Date	2026-10-04
Author	Amish Chadha	License	CERN-OHL-S-2.0
Project TRL	TRL 3 Proof of concept		
Repository	github.com/BoujeeEnjinia1701/sinkgrab		
Source	docs/03-requirements.md @ uncommitted		

REVISION HISTORY

Version	Date	Author	Change
0.5	2026-10-04	Amish Chadha	Amish's round-3 decision 5A of 2026-10-04 ("For round 3, I agree with all your proposed recommendations"), SKG-DDR-004: slotted sleeve and cross pin give the scraper's arms the 300 mm of sleeve travel they need to fold; status of R2, R9 and R11 from SKG-CAL-001 v0.3; no requirement restated
0.4	2026-10-03	Amish Chadha	Amish's decisions 27A, 28B and 29B of 2026-10-03 ("i agree with all the 46 recommendations you provided. please proceed."), SKG-DDR-003: R4 restated for three people at the cranks during the hoist; status of R1, R3, R4, R5, R7, R9 and R11 from SKG-CAL-001 v0.2
0.3	2026-10-03	Amish Chadha	TRL 3 status from SKG-CAL-001 on the constructable design (SKG-DDR-002); R1 and R3 at risk and R4 not met, each posed to Amish as a decision in SKG-DEC-001
0.2	2026-10-03	Amish Chadha	

TRL 2 requirements with concept status (SKG-DDR-001); R11 restated against the value-engineering target

0.1	2026-09-30	Amish Chadha	Initial scaffold
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SinkGrab requirements

Eight of the eleven requirements are met on paper or by design, one can only be shown in a field trial, one is at risk and one is not met on paper by a small margin (SKG-CAL-001 v0.2, section M). Amish decided the three open requirements on 2026-10-03 ("i agree with all the 46 recommendations you provided. please proceed."; SKG-DDR-003): a third person at the cranks during the hoist (R4, restated below), a 3:1 closing tackle (R3) and toe blades that cut 40 mm past the ring's outer face (R1). The cycle time (R4) is now about 3.2 min at 10 m, 6 % over the target; the timed trial decides it. The depth gained (R1) still depends on how much the soil grips the sinking rings, which only the first trial can measure. On 2026-10-04 Amish decided the scraper sleeve's travel ("For round 3, I agree with all your proposed recommendations"; SKG-DDR-004): a 340 mm slotted sleeve on a 12 mm cross pin, so the arms fold to pass the ring and R2 is met by design as stated. "Met on paper" means shown by calculation, not by test.

Safety: SinkGrab is lifting equipment worked over an open well shaft. R7 (proof load) and R8 (nobody in the well) are its safety requirements, and the overload limits behind R7 (a crank shear pin and a shear link, SKG-CAL-001, section C) are checked on paper only. Every lifting part must be proof-loaded before use (SKG-BLD-001, section 6).

Table 1. Requirements and status at TRL 3

ID	Requirement	Target	Verification (TRL 3 or later)	Status (SKG-CAL-001)
R1	Deepen a flooded caisson well below the water table without de-watering	At least 3 m of standing water depth reached (target)	Field trial in a partner well, measured with the dip tape before and after	At risk: the 8-ring string sinks only while skin friction stays below about 1.7 kPa; estimated 1 to 3 kPa; the toes now cut 40 mm past the ring's outer face to keep it low (decision 29B); friction measured in the first trial
R2	Fit common caisson ring sizes	Works inside rings of 0.8 to 1.3 m internal diameter	Fit check in rings of three sizes	Met by design: grab 468 mm across; scraper folds to 389 mm radius on 300 mm of sleeve travel (slotted sleeve and cross pin, SKG-DDR-004); toes, skids and a longer arm pair cover the range
R3	Grab load per cycle	At least 20 L of saturated sand per bite (target)	Test pit trials with a measured tipping bin	Met on paper at the assumed fill: 21.2 L at 75 % fill, closed by a 3:1 tackle with 50 % more lip force (decision 28B); 14.1 L at 50 %; fill confirmed in the test-pit trial
R4	Cycle time	3 min or less per grab cycle at 10 m depth with three people at the cranks during the hoist and two otherwise (target; restated 2026-10-03, decision 27A)	Timed trial over 20 cycles	Not met on paper by about 0.2 min: 3.17 min with three at the hoist (3.8 min with two); the timed trial decides
R5	Operator effort	Crank force 150 N or less per operator at rated grab load (target)	Spring balance on the crank handle during lifts	Met on paper: 54 N each with two people (108 N for one)
R6	Keep the lining vertical while sinking	Tilt no worse than 1 in 80, the allowance used for sunk well foundations (Construction Civil)	Four-point dip tape readings every cycle during trials	Not verifiable at TRL 3: readings resolve about 1 in 250; correction by scraping the high side and saddles
R7	Proof load of lifting parts	Tripod, capstan, hooks and rigging proof-loaded to 2 times working load (target)	CalRig proof-load test with record	Met on paper: proof 193 kg, inside the HatchSide proof of 225 kg; line factor 10.4; shear link margin 1.15
R8	No person in the well during digging	100% of digging and undercutting done from the surface	Trial log and method statement review	Met by design: every task from the surface; doors cover the shaft
R9	Portability	Heaviest single piece 25 kg or less; kit fits in a small pickup (target)	Weigh each piece; load trial	Met on paper: heaviest piece 22.9 kg (capstan frame) with the scraper arms carried unpinned (scraper head 26.1 kg with them on, 20.1 kg without); kit about 457 kg, longest piece 2.0 m
R10	Local build	Built with a stick welder, drill and hand tools from common steel section	Build by a partner workshop from the published drawings	Met by design; the shell skins need plate rolls, which most fabricators have

ID	Requirement	Target	Verification (TRL 3 or later)	Status (SKG-CAL-001)
R11	Prototype cost	At or under the USD 4,000 value-engineering target (a hypothetical control target)	Bill of materials and receipts	Met on paper, within the target: USD 2,566 (USD 1,434 under)

■ Requirements at risk or not met

Amish decided each of these on 2026-10-03 ("i agree with all the 46 recommendations you provided. please proceed."); the records are SKG-DDR-003 and SKG-DEC-001.

- **R4 (not met on paper by about 0.2 min).** Restated by decision 27A: the target is 3 min a cycle at 10 m with three people at the cranks during the hoist. Hand power still sets the hoist: 76 s of the 3.17 min cycle. The timed trial over 20 cycles decides.
- **R3 (met on paper at the assumed fill).** Decision 28B: a 3:1 tackle closes the lips 50 % harder (about 685 N). The fill is confirmed in the test-pit trial.
- **R1 (at risk).** Decision 29B: toe blades cut 40 mm past the ring's outer face to loosen the soil against it. Sinking still stops if skin friction exceeds about 1.7 kPa; the first trial measures it.

■ Assumptions

- Crews already use precast caisson rings of a smaller diameter below the water table; the design case is a 1.0 m ring with a 75 mm wall inside a 1.3 m lining.
- Soils at the target sites can be taken by a clamshell grab; boulders and hard rock need a separate method.
- The HatchSide tripod keeps its 150 kg rating for material handling; SinkGrab's overload limits keep the line inside it.
- Three people are available to work the capstan for a full day (two between hoists, three during the hoist); a crew of four or five works the site.